

# fwhFoam: A Ffowcs Williams–Hawkings solver with on-surface acoustic source localization for OpenFOAM

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## Abstract

We present **fwhFoam**, an open-source aeroacoustic post-processing library for the OpenFOAM computational fluid dynamics toolbox that couples far-field noise prediction with *on-surface acoustic source localization*. Far-field sound is computed with Farassat’s formulation 1A of the Ffowcs Williams–Hawkings (FW-H) equation, evaluated by a source-time-dominant (advanced-time) algorithm, for permeable and impermeable integration surfaces through one unified code path and with a uniform mean flow handled in closed form via the Garrick-triangle emission-time relation. The distinguishing feature is a built-in implementation of the surface sound-power density  $\sigma$  derived from the diffraction-filter theory of Delfs & Ruck: from the same permeable-surface data, **fwhFoam** produces a map of *where on the surface the radiated sound originates*, with an exact power identity  $\oint \sigma dS = P_{\text{far-field}}$  serving as an internal consistency check. To our knowledge no other OpenFOAM FW-H tool offers this. The software comprises a runtime function object, an offline reprocessing utility, a Python companion package with the localization module, and verification suites. Far-field predictions match analytic monopole, dipole and convected-monopole solutions to  $\sim 0.1\%$  with second-order convergence, and the  $\sigma$  power identity is verified to a few tenths of a percent on analytic sources.

**Keywords:** Aeroacoustics, Ffowcs Williams–Hawkings, Acoustic source localization, Surface sound-power density, OpenFOAM, Computational aeroacoustics

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## Highlights

- Open-source Ffowcs Williams–Hawkins acoustic solver for OpenFOAM.
- Permeable and impermeable surfaces via one code path; closed-form mean flow.
- Unique on-surface source localization by a surface sound-power density map.
- A built-in identity ties the source map to the far-field radiated power.
- Verified to  $\sim 0.1\%$  on analytic sources with second-order convergence.

## Code metadata

| Code metadata description              | Value                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------|
| Current code version                   | v1.0.0                                                                                          |
| Permanent link to code/repository      | <a href="https://github.com/Sparsh-Sharma/fwHfoam">https://github.com/Sparsh-Sharma/fwHfoam</a> |
| Legal code license                     | GPL-3.0-or-later                                                                                |
| Code versioning system used            | git                                                                                             |
| Software code languages, tools         | C++ (C++14), Python 3, OpenFOAM (wmake)                                                         |
| Compilation requirements, dependencies | OpenFOAM v2306; C++14 compiler; Python $\geq 3.9$ with NumPy                                    |
| Developer documentation                | README.md, doc/theory.md, doc/fwH-data-format.md                                                |
| Support email for questions            | sparsh.sharma@dlr.de                                                                            |

## 1. Motivation and significance

Aerodynamically generated sound is a central concern in the design of aircraft, rotors, wind turbines, ground vehicles and cooling fans. Direct computation of the acoustic field together with the flow is prohibitively expensive for most engineering configurations, because the acoustic fluctuations are many orders of magnitude smaller than the hydrodynamic ones and the far field lies many wavelengths from the source. The established remedy is an *acoustic analogy*: the near-field unsteady flow is computed by scale-resolving

CFD (LES, DES, DNS or a lattice Boltzmann method), and an integral formulation propagates the sound to arbitrary far-field observers at negligible additional cost.

The Ffowcs Williams–Hawkings (FW-H) equation [1] is the most general such analogy, valid for surfaces in arbitrary motion and, in its permeable (porous-surface) form [2, 3], able to account for volume (quadrupole) sources by enclosing them within an off-body integration surface. Farassat’s formulation 1A [4] is the standard time-domain realisation and underpins most rotorcraft and turbomachinery noise prediction tools.

Mature open-source FW-H implementations for OpenFOAM already exist; the most established is the `libAcoustics` library [6], which provides Curle, Farassat 1A (permeable and impermeable) and a wind-tunnel formulation. `fwhFoam` is not intended to replace or compete with such tools on the far-field prediction alone, and we make no claim of novelty for the FW-H formulation itself, which is standard. Its contribution is twofold. First, it makes a compact, independently verifiable FW-H capability from our institution openly available to the OpenFOAM community, with a bundled analytic verification and convergence suite that any user can reproduce. Second, and distinctively, it couples the far-field prediction with *on-surface acoustic source localization*: the same permeable-surface data used for the FW-H integral are processed into a surface sound-power density  $\sigma(\mathbf{x})$  that identifies which parts of the surface radiate to the far field. This quantity, and the diffraction-filter theory it derives from [7], addresses a question the far-field signal cannot answer — localization of the true acoustic sources on a surface — and is, to our knowledge, not available in any existing OpenFOAM FW-H tool. A built-in power identity,  $\oint \sigma dS = P_{\text{far-field}}$ , ties the two together and provides an internal consistency check.

## 2. Software description

### 2.1. Architecture

`fwhFoam` is organised around a mesh-independent formulation engine that is shared by two front-ends:

- `libfwhFoam` — a shared library providing the class `fwhFormulation1A` (the numerics) and a runtime OpenFOAM *function object* `fwh` that samples surface data during a simulation;

- **fwhSolve** — a stand-alone utility that recomputes observer signals from stored surface data, allowing observers, ambient conditions or the mean-flow speed to be changed without re-running the CFD;
- **pyfwh** — a Python package for reading and writing the surface-data format, generating analytic reference fields and integration surfaces, computing spectra, and — through its **localization** module — evaluating the surface sound-power density  $\sigma$ ;
- **sigma\_localize** — a driver that turns permeable surface-data files into a  $\sigma$  map (VTK/CSV) and reports the power identity.

The formulation engine operates purely on raw face-centre data (position, unit normal, area, and per-time-step samples of gauge pressure, density and fluid velocity), so identical numerics serve the patch-based (impermeable) and sampled-surface (permeable) paths and the runtime and offline front-ends.

## 2.2. Governing equations

The acoustic pressure is split into thickness ( $T$ ) and loading ( $L$ ) contributions. For a static integration surface  $S$  with outward unit normal  $\hat{n}$  in a uniform mean flow of Mach vector  $\mathbf{M} = -\mathbf{U}_0/c_0$ ,

$$4\pi p'_T(\mathbf{x}, t) = \int_S \left[ \frac{\rho_0 \dot{U}_n}{r(1 - M_r)^2} + \frac{\rho_0 U_n c_0 (M_r - M^2)}{r^2(1 - M_r)^3} \right]_{\text{ret}} dS, \quad (1)$$

$$4\pi p'_L(\mathbf{x}, t) = \int_S \left[ \frac{\dot{L}_r}{c_0 r(1 - M_r)^2} + \frac{L_r - L_M}{r^2(1 - M_r)^2} + \frac{L_r c_0 (M_r - M^2)}{r^2(1 - M_r)^3} \right]_{\text{ret}} dS, \quad (2)$$

with the permeable-surface source vectors

$$U_i = v_i + \frac{\rho}{\rho_0}(u_i - v_i), \quad L_i = p' \hat{n}_i + \rho u_i(u_n - v_n), \quad (3)$$

where  $u$  is the fluid velocity,  $v$  the surface velocity,  $r$  and  $\hat{r}$  the source-to-observer distance and unit vector,  $M_r = \mathbf{M} \cdot \hat{r}$ ,  $L_M = \mathbf{L} \cdot \mathbf{M}$ , subscript  $n$  a projection on  $\hat{n}$ , subscript  $r$  on  $\hat{r}$ , and a dot a source-time derivative. For a solid wall  $u_n = v_n$  and Eqs. (3) reduce to the classical impermeable form, so no separate code path is required.

### 2.3. Advanced-time algorithm

Rather than solving the implicit retarded-time equation, `fwhFoam` marches in source time: an element emitting at time  $\tau$  is received at  $t = \tau + T$ , where the emission-to-reception delay  $T$  follows from the closed-form Garrick-triangle relation

$$(c_0^2 - U_0^2) T^2 + 2(\mathbf{d} \cdot \mathbf{U}_0) T - |\mathbf{d}|^2 = 0, \quad \mathbf{d} = \mathbf{x}_{\text{obs}} - \mathbf{y}, \quad (4)$$

requiring only  $|\mathbf{U}_0| < c_0$ . Each element's contribution is deposited onto a uniform observer-time grid by exact piecewise-linear resampling between consecutive (Doppler-shifted) arrival times; time derivatives use second-order central differences over three buffered source levels. The scheme is single-pass and streaming (three source levels in memory) and parallelises by having each MPI rank integrate its own faces, the partial observer signals being combined by a fixed-size reduction.

### 2.4. On-surface source localization

The distinguishing capability operates on the permeable-surface Cauchy data — the surface pressure  $p$  and normal velocity  $v_n = \mathbf{u} \cdot \hat{n}$  that the FW-H surface already samples. Working per angular frequency  $\omega = c_0 k$  with the complex spectra  $\phi = \hat{p}$  and  $\psi = \partial \hat{p} / \partial n = -i\omega \rho_0 \hat{v}_n$ , the radiating field has far-field directivity  $F(\hat{\mathbf{x}}) = \frac{1}{4\pi} \oint [ik(\hat{\mathbf{x}} \cdot \hat{n})\phi - \psi] e^{ik\hat{\mathbf{x}} \cdot \boldsymbol{\xi}} dS$  and radiated power  $P = \frac{1}{2\rho_0 c_0} \oint |F|^2 d\Omega$ . Carrying out the angular integration in closed form yields an exact surface sound-power density

$$\sigma(\boldsymbol{\xi}) = \frac{1}{32\pi^2 \rho_0 c_0} \text{Re} \oint \left[ k^2 (\hat{n} \mathbf{W} \hat{n}') \phi \phi'^* - ik(\hat{n} \mathbf{V}) \phi \psi'^* + ik(\hat{n}' \mathbf{V}) \psi \phi'^* + J_0 \psi \psi'^* \right] dS(\boldsymbol{\xi}'), \quad (5)$$

with closed-form kernels  $W_{ij} = 4\pi[\delta_{ij} j_1(z)/z - \hat{d}_i \hat{d}_j j_2(z)]$ ,  $V_i = 4\pi i j_1(z) \hat{d}_i$ ,  $J_0 = 4\pi j_0(z)$ , where  $\mathbf{d} = \boldsymbol{\xi} - \boldsymbol{\xi}'$ ,  $z = k|\mathbf{d}|$ ,  $\hat{d} = \mathbf{d}/|\mathbf{d}|$ . By construction  $\oint \sigma dS = P$ , so the localization map and the far-field power are two views of the same data and must agree. On a rigid wall ( $\psi = 0$ ) only the pressure block survives, recovering the classical rigid-surface density [7]. The density is signed, reflecting that radiated power is a two-point quantity.

### 2.5. Usage

The runtime function object is enabled from `system/controlDict`; switching between an impermeable and a permeable surface is a change of a single sub-dictionary:

```

fwh1
{
    type fwh; libs (fwhFoam);
    c0 340.29; rho0 1.225; U0 (0 0 0);
    surface { type patches; patches (cylinder); } // impermeable
    observers { mic1 { position (0 10 0); } }
    executeControl timeStep; executeInterval 8;
}

```

### 3. Illustrative examples

#### 3.1. Analytic verification

The distributed test suite feeds exact acoustic fields onto a geodesic icosphere enclosing the sources and compares the predicted observer pressure with the analytic far field. Because the surface data are exact, a correct formulation must reproduce the analytic signal to discretisation accuracy. Table 1 lists the relative  $L_2$  error for three canonical cases, isolating the thickness term (monopole), the loading term (dipole) and the convective/Garrick terms (convected monopole at Mach 0.3). All radiating directions agree to within about 0.1 %.

Table 1: Relative  $L_2$  error of **fwhFoam** against analytic far-field solutions (icosphere surface, subdivision level 3, 1280 faces, 64 samples per period).

| Case                          | Term exercised                | Relative $L_2$ error                      |
|-------------------------------|-------------------------------|-------------------------------------------|
| Monopole (medium at rest)     | thickness $\dot{U}_n$         | $1.6 \times 10^{-3}$                      |
| Dipole (medium at rest)       | loading $L_r, \dot{L}_r$      | $7 \times 10^{-4}$ – $1.6 \times 10^{-3}$ |
| Convected monopole, $M = 0.3$ | Garrick delay + $(M_r - M^2)$ | $1.6 \times 10^{-3}$                      |

#### 3.2. Convergence

Figure 1 shows the error of the monopole case as the surface resolution and the time step are refined. Both sequences exhibit second-order convergence (observed orders  $p \approx 2.6$  in space and  $q \approx 2.3$  in time), consistent with midpoint-rule surface quadrature and central-difference time derivatives applied to a smooth field.

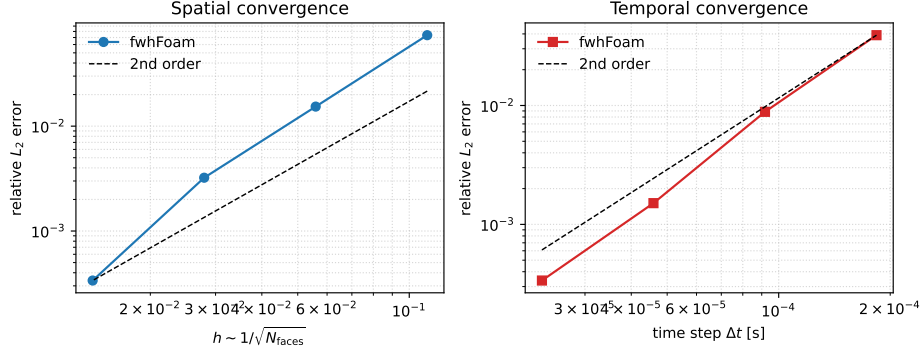


Figure 1: Spatial (left) and temporal (right) convergence of the relative  $L_2$  error for the analytic monopole. Dashed lines indicate second-order slopes.

### 3.3. Verification of the source-localization density

The power identity underlying  $\sigma$  is verified independently of the FW-H solver. Exact monopole and dipole fields are placed on a geodesic icosphere enclosing the source, and three quantities are compared: the analytic radiated power (from the surface acoustic intensity of the exact field), the far-field power from the directivity  $F$ , and the integral  $\oint \sigma dS$  of the two-point density (5). For 1280 faces they agree to  $2.9 \times 10^{-3}$  (relative), and  $\sigma$  reproduces the correct spatial structure: uniform over the sphere for the isotropic monopole, and vanishing at the poles while peaking at the equator for the dipole — the  $\cos^2$  radiation pattern. This confirms both the magnitude and the localization content of the density.

### 3.4. Aeolian tone of a circular cylinder

The tutorial `cylinderAeolianTone` computes laminar flow past a 2D circular cylinder at  $Re = 100$  with the incompressible `pimpleFoam` solver and applies the runtime `fwh` function object to the cylinder wall (impermeable surface). Vortex shedding produces a fluctuating lift and hence a dipole tone at the Strouhal frequency. Figure 2 shows the predicted far-field pressure spectrum, whose fundamental coincides with the shedding frequency inferred from the lift coefficient, and the characteristic dipole directivity that peaks normal to the mean flow.

The same simulation, sampled on a permeable cylindrical control surface of radius  $8D$ , yields the source-localization density (5) at the shedding tone (Fig. 3). The density is concentrated on the downstream sector of the control

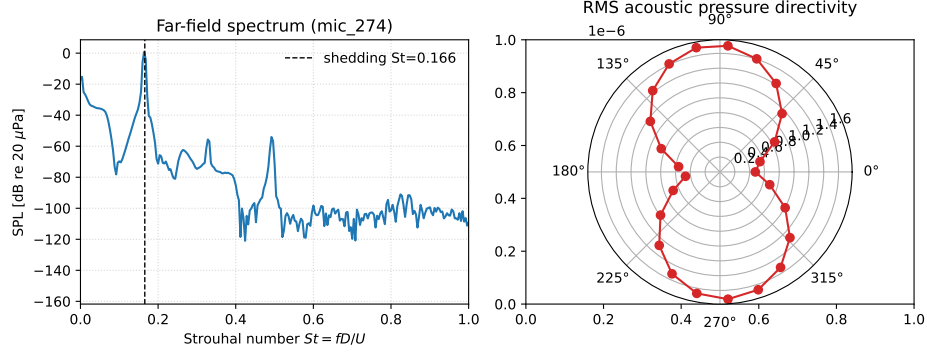


Figure 2: Aeolian tone of a cylinder at  $Re = 100$ : (left) far-field pressure spectrum at the cross-flow observer, showing the tone at the shedding Strouhal number; (right) root-mean-square acoustic pressure directivity, exhibiting the lift-dipole lobe pattern.

surface, where the unsteady wake crosses it, and forms a signed lobe pair — reflecting that radiated power is a two-point rather than a pointwise quantity. Its surface integral reproduces the independently evaluated far-field power to machine precision, the built-in identity confirming the internal consistency of the computation on genuine CFD data. The map thus attributes the tonal radiation to a localized region of the surface rather than to the surface as a whole.

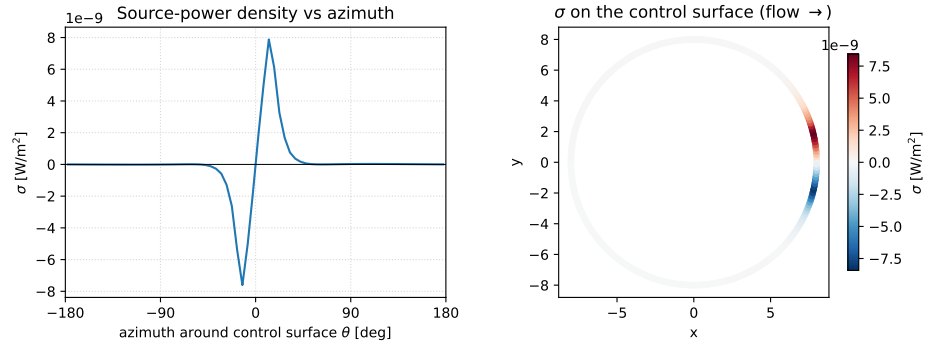


Figure 3: Source-power density  $\sigma$  on the permeable control surface ( $r = 8D$ ) at the shedding tone: (left) azimuthal profile around the surface; (right)  $\sigma$  over the surface (mean flow left to right). The density localizes to the downstream wake-crossing sector and is signed, integrating to the far-field radiated power  $\oint \sigma dS = P$  to machine precision.

## 4. Impact

`fwhFoam` contributes to the OpenFOAM aeroacoustics ecosystem in two ways. As a far-field solver it provides a compact, documented and independently verified FW-H capability that complements existing libraries; the offline `fwhSolve` utility decouples the expensive CFD from the inexpensive acoustic post-processing, so observer arrays, ambient conditions and mean-flow speeds can be explored without re-running the flow solver. Its more significant impact is to bring *acoustic source localization* into a standard CFD workflow: the surface sound-power density  $\sigma$  turns an FW-H integration surface, normally used only to obtain a far-field signal, into a diagnostic that shows which surface regions actually radiate. This directly supports noise-reduction design — identifying the parts of an airframe, edge or porous treatment responsible for the radiated sound — which a far-field spectrum alone cannot. Because  $\sigma$  is computed from the same permeable-surface data as the FW-H integral and carries a built-in consistency identity, it adds this capability at negligible marginal cost and with a self-check. The bundled analytic verification and convergence suites provide a template for reproducible validation, and the mesh-independent formulation engine is a foundation for moving-surface (rotating-frame) and frequency-domain extensions.

## 5. Conclusions

We have described `fwhFoam`, a compact and independently verifiable Ffowcs Williams–Hawkings solver for OpenFOAM supporting permeable and impermeable integration surfaces and a uniform mean flow, whose distinguishing feature is a built-in on-surface acoustic source localization: the surface sound-power density  $\sigma$ , computed from the same data as the far-field integral, with an exact power identity as an internal check. The far-field solver is verified against analytic monopole, dipole and convected-monopole solutions to about 0.1% with second-order convergence, and the localization density is verified through its power identity on analytic sources to a few tenths of a percent. Planned extensions include moving and rotating integration surfaces, the diffraction-filter projection for rigid-wall data, and a frequency-domain formulation.

## Acknowledgements

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